IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-44

Transformer Design For Isolated DC-DC Converter-I

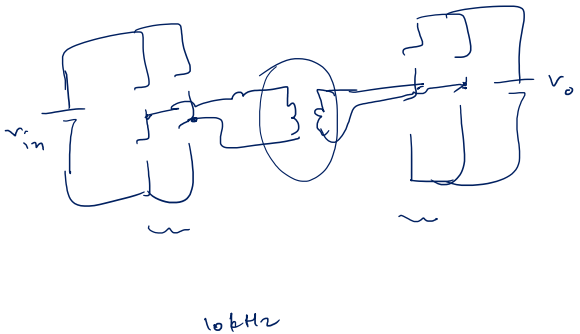
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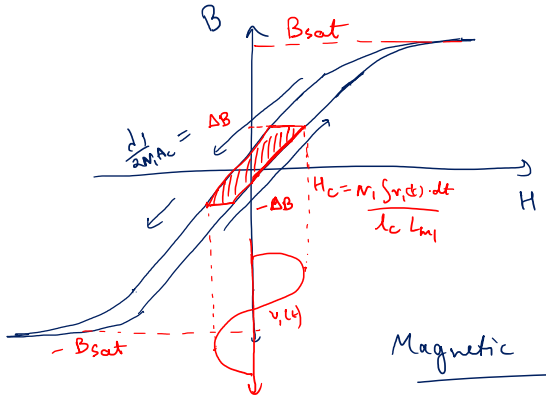
Recap

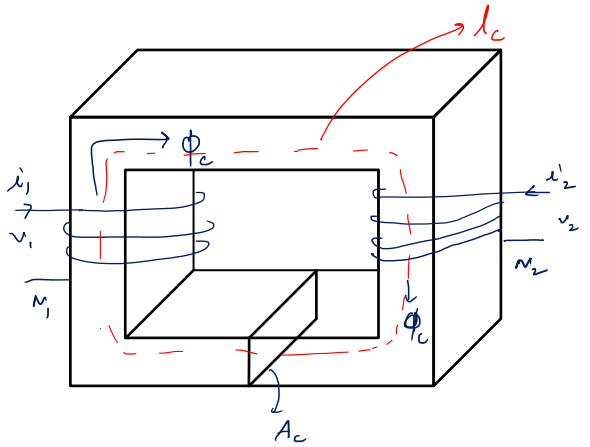
- Full- bridge converter : PWM, PSFB
- Dual Active Bridge



Transformer

- Consists of core and winding





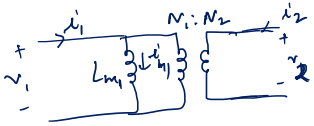
from the faraday's law

$\Rightarrow v_1(t) = -N_1 \frac{d\phi_c}{dt}$

$\Rightarrow \int v_1(t)\,dt = N_1 \phi_c = \lambda_1$ → flux linkage

$\Rightarrow$

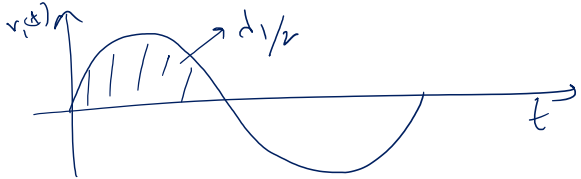
Magnetic circuit model ← Transformer model



$\Rightarrow H_c = \frac{N_1 i_{m1}}{l_c}$ (Ampere's circuital law)

$\Rightarrow i_{m1} = \frac{1}{L_{m1}} \int v_1(t)\,dt$

Transformer



$$\Rightarrow \quad v_1(t) = N_1 \frac{d\phi_c}{dt}$$

$$\Rightarrow \quad v_1(t) = N_1 A_c \frac{dB_c}{dt}$$

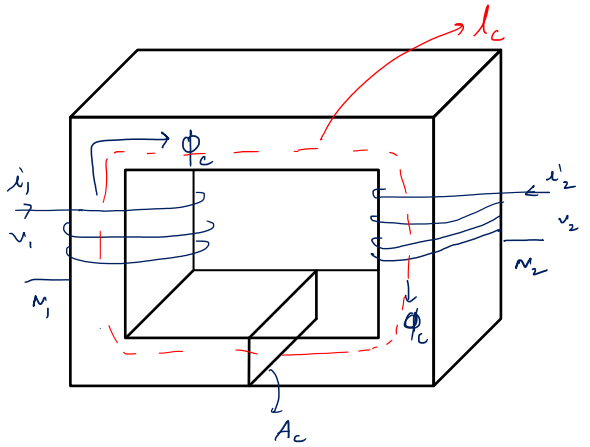
$$\Rightarrow \quad \int v_1(t)\,dt = N_1 A_c \cdot \Delta B$$

$$\Rightarrow \quad d_1 = 2 N_1 A_c \Delta B$$

$$\Rightarrow \quad \Delta B = \frac{d_1}{2 N_1 A_c}$$

$B_c \rightarrow$ Magnetic flux density

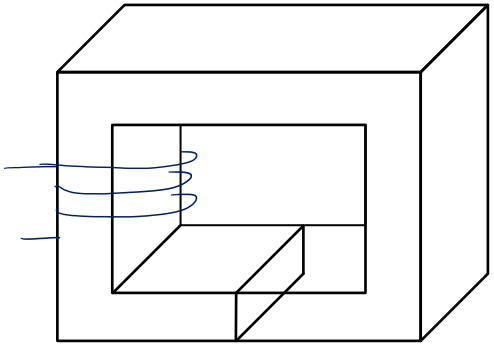
$B_c = \phi_c \cdot A_c$



Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Transformer

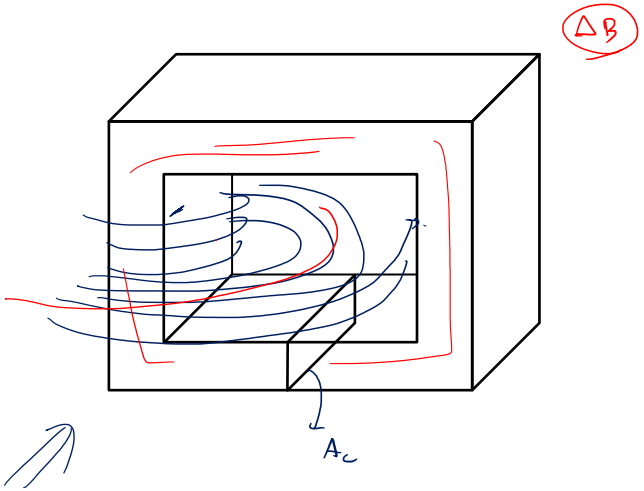
- The core loss and winding loss are significant
- Need to find optimum flux density to reduce both losses



Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Transformer Design

- Design Outcome
  - Core size (core cross-sectional area, window area, mean length per turn, magnetic path length)
  - Number of turns
  - Wire size
  - Peak flux density



Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Core Loss

- The core loss comprises of Hysteresis loss and eddy current loss
- Hysteresis loss is due to the energy require to change the magnetization of the core.
- Eddy current loss is due to currents induced in the core

→ rotation of magnetic domain

→ Core materials are conductive in nature

Hysteresis loss.

The energy going into the core (via winding) in one cycle

$$W = \int_{\text{one cycle}} v_1(t) \cdot i_1(t) \cdot dt \longrightarrow (1)$$

from faraday's law $\Rightarrow$ $v_1(t) = \frac{d\lambda_1}{dt} = N_1 A_c \frac{dB}{dt}$

from Ampere's law $\Rightarrow$ $H l_c = N_1 i_1(t)$

$\Rightarrow$ $i_1(t) = \frac{H l_c}{N_1}$

$\Rightarrow$ $W = \int_{\text{one cycle}} N_1 A_c \frac{dB}{dt} \cdot \frac{H l_c}{N_1} \cdot dt = A_c l_c \int_{\text{one cycle}} H \cdot dB$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

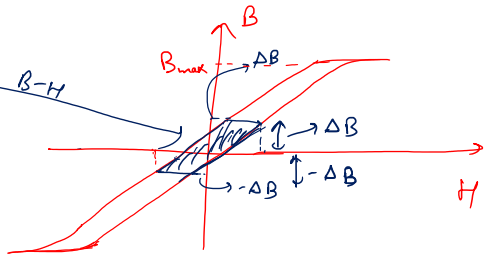
Core Loss Cont.

Hysteresis Power loss, $P_{Hys} = f \, A_c l_c \left(\int_{\text{one cycle}} H \cdot dB \right)$

$A_c l_c$ → volume of the core

$\int_{\text{one cycle}} H \cdot dB$ → Area enclosed within the B-H curve

f → excitation frequency



$\int_{\text{one cycle}} H \cdot dB$ → area of B-H loop enclosed during one cycle of operation

$P_{Hys} \propto f (\Delta B)^{\alpha} A_c l_c$

ΔB → the peak value of ac flux density (from the average value)

α → constant term

B-H curve is not changing for the particular excitation frequency

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Core Loss Cont.

- Eddy current also increases with excitation frequency

→ resistive loss

If core material has resistive impedance → R_{core}

$\Rightarrow P_{eddy} \propto i_{eddy}^2 R_{core}$

$\Rightarrow i_{eddy} \propto v_{induced}$

$v_{induced} \propto (\Delta\Phi)$
$v_{induced} \propto f$
$\Rightarrow i_{eddy} \propto \Delta\Phi$, $i_{eddy} \propto f$

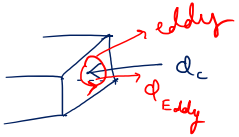
$\Rightarrow P_{eddy} \propto f^2 (\Delta\Phi)^2 \cdot R_{core}$

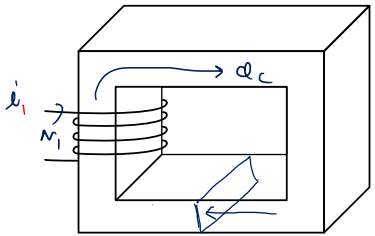
$\Rightarrow P_{eddy} \propto f^2 (\Delta B)^2 A_c^2 R_{core}$

$\Rightarrow P_{eddy} \propto f^2 (\Delta B)^2 A_c l_c$

($R_{core} \propto l_c$, $\propto 1/A_c$)

$A_c l_c \rightarrow$ volume of the core





Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Core Loss Cont.

$P_{core} = K_{core} (\Delta B)^{\beta} (A_c l_c)$ → for a given excitation frequency (sinusoidal)

$A_c l_c$ → volume of the core

→ empirical steinmetz formula

K_{core} → Constant of proportionality and it depends on excitation frequency

β → exponent which is obtained from manufacturer's datasheet

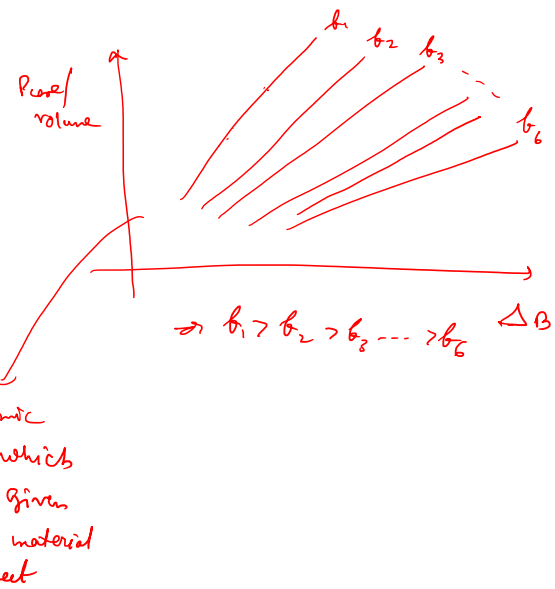
→ Material property

$\beta \rightarrow 2.5$ to 2.7 (for ferrite material)

$$\Rightarrow \frac{P_{core}}{Volume} = K_{core} (\Delta B)^{\beta}$$

→ peak value of AC flux density

$$\Rightarrow P_{core} \propto (\Delta B)^{\beta}$$



logarithmic curve, which is being given in core material datasheet

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Maximum Flux Density

From Faraday's law of electromagnetism

$$\lambda_1 = \int v_1(t)\,dt$$

$v_1(t)$ → arbitary periodic input given to the primary winding

⇒ the peak value of ac flux density

$$\Delta B = \frac{\lambda_1}{2 n_1 A_c}$$

λ_1 → flux linkage in winding 1

A_c → Area of cross section of core

n_1 → no. of turns in the primary winding or winding 1

$$\Delta B < B_{sat}$$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Thank You